

André Anjos

HEAD OF THE MEDICAL ARTIFICIAL INTELLIGENCE GROUP, PH.D.



About me

André Anjos leads the Medical AI Group at the Idiap Research Institute, working on machine learning, computer vision, and signal processing for medical decision making, research reproducibility, and open science.

Interests

- Pattern Recognition
- Machine Learning
- Reproducible Research
- Computer Vision
- Signal Processing
- Artificial Intelligence
- Medical AI

Contact

✉ andre.anjos@idiap.ch

☎ +41 27 721 7763

🏠 Idiap Research Institute,
rue Marconi 19, CH-1920
Martigny, Switzerland

Personal

Date of birth: 31.12.1973 (52 years old)

Nationality: Swiss and Brazilian

Family: Married, one child

🌐 anjos.ai

🔗 anjos

🌐 andreranjos

🏠 pAflhMoAAAAJ

🆔 0000-0001-7248-4014

🏠 medai

PROFESSIONAL EXPERIENCE

- 2018 – present **Scientific Researcher**
Idiap Research Institute
Head of the Medical Artificial Intelligence Group. 📍 Martigny, Switzerland
- 2019 – 2023 **Lecturer**
Idiap Research Institute & Unidistance Suisse
Three modules of the Master in Artificial Intelligence: open science and ethics (M05), and fundamentals of machine learning (M06 and M08). 📍 Martigny, Switzerland
- 2013 – 2023 **Lecturer**
École Polytechnique Fédérale de Lausanne
EE-612, Fundamentals of Statistical Pattern Recognition: a doctoral course on machine learning and pattern recognition, from the basics to deep learning, given biennially. 📍 Lausanne, Switzerland
- 2014 – 2018 **Research Associate**
Idiap Research Institute
Research on biometrics, security and computing. 📍 Martigny, Switzerland
- 2010 – 2013 **Post-doctoral Researcher**
Idiap Research Institute
Research on biometrics, security and computing. 📍 Martigny, Switzerland
- 2004 – 2010 **Research & Development Engineer**
University of Wisconsin, Madison
• Design, development and maintenance of the ATLAS Trigger and Data Acquisition systems, deployed and operational on more than 5000 machines at CERN and worldwide. 📍 CERN, Switzerland
- 2000 – 2004 **Research Assistant**
Federal University of Rio de Janeiro
Design, implementation and integration of a novel algorithm for fast physics triggering — based on calorimetry, topological mapping and neural networks — into the second-level trigger software of the ATLAS Experiment, alongside the online infrastructure needed to run and validate it. 📍 CERN, Switzerland
- 1997 – 1998 **Research Internship**
CERN
Contributed to early prototypes of the second-level trigger and to what became the final trigger design of the ATLAS Experiment. 📍 CERN, Switzerland
- 1994 – 1999 **Research Internship**
Federal University of Rio de Janeiro
• Sub-optimal filtering for particle discrimination.
• Neural classifiers implemented in a transputer-based parallel machine. 📍 Rio de Janeiro, Brazil

EDUCATION

- 2001 – 2006 **Ph.D., Electronics Engineering (Signal Processing & Machine Learning)**
Federal University of Rio de Janeiro
Electronics Engineering School (Signal Processing Laboratory), Post-Graduate Program (COPPE). Thesis developed at CERN, Switzerland, in the context of the ATLAS Experiment. 📍 Rio de Janeiro, Brazil
- 2000 – 2001 **M.Sc., Electronics Engineering (Signal Processing & Machine Learning)**
Federal University of Rio de Janeiro
Electronics Engineering School (Signal Processing Laboratory), Post-Graduate Program (COPPE). Thesis developed in the context of the ATLAS-CERN / UFRJ collaboration. 📍 Rio de Janeiro, Brazil
- 1994 – 1999 **Engineer, Electronics Engineering**
Federal University of Rio de Janeiro
Degree from the Electronics Engineering Department. 📍 Rio de Janeiro, Brazil
- 1989 – 1993 **Technician, Electronics**
Federal Center of Technology (CEFET)
Technical course in electronics, taken alongside high-school studies. 📍 Rio de Janeiro, Brazil

RESEARCH AREAS

- 2023–2026 **Ophthalmology — Uveitis**

Languages

Portuguese	Mother tongue
English	Fluent
French	Fluent, certified C1
Spanish	Basic, reads well

Computer Skills

Linux, macOS, Windows
System and network administration
Structured and object-oriented languages
Concurrent and parallel programming
Large software projects, releases and policies
Revision control, CI, documentation
Database design and use

Hôpital ophtalmique Jules-Gonin (HOJG), Lausanne; University Hospital of Grenoble Alpes; Luzerner Kantonsspital

AI-assisted grading of intraocular inflammation in uveitis from retinal fluorescein angiography, developed with the Jules-Gonin Eye Hospital.

anjos.ai/projects/uveitis

2017 Neurophysiology — Sleep Medicine

Generalisable, reproducible automatic sleep-stage classification from polysomnography, working toward foundation models for sleep medicine.

anjos.ai/projects/sleep-medicine

2023–2026 Trustworthy and Fair Medical AI

Federal University of São Paulo (UNIFESP)

Making medical AI fair across patient subgroups and useful in the clinic, with evaluation frameworks, foundation-model bias analysis, and utility-fairness trade-offs.

anjos.ai/projects/trustworthy-fair-ai

2020–2024 Ophthalmology — Retinal Image Analysis

University Hospital of Grenoble Alpes; Idiap Research Institute

Segmentation and analysis of retinal fundus images (vessels, optic disc, blood flow) with minimalistic, reproducible, and clinically deployable models.

anjos.ai/projects/retinal-image-analysis

2021–2024 Radiology — Robust Image Analysis

Federal University of Rio de Janeiro; University of Zürich; Murabei Data Science, Brazil; University Hospital Basel; Lausanne University Hospital (CHUV); University of Geneva; ETH Zurich

Interpretable, generalisable computer-aided analysis of radiological images — detecting tuberculosis on chest X-rays and extracting stable quantitative biomarkers from CT — built to hold up across scanners and populations.

anjos.ai/projects/tuberculosis-radiology

2020–2025 Epidemiology — Anticipating Risk

University of Geneva; Polytechnic University of Leiria; Federal University of Rio de Janeiro; State University of Campinas (UNICAMP); Federal University of Lavras

Machine learning on routine clinical records, chest radiographs, and patient-contact networks to flag who is most at risk — from diabetic-foot complications to drug-resistant infection to tuberculosis.

anjos.ai/projects/tuberculosis-epidemiology

Angiology — Fibromuscular Dysplasia

CHUV, Lausanne University Hospital

Machine learning for the characterisation and diagnosis of fibromuscular dysplasia, in collaboration with CHUV's angiology department.

anjos.ai/projects/fibromuscular-dysplasia

2015–2026 Reproducible Research and Open Science

Software suites and open platforms that make computational research repeatable, shareable, extensible, and stable, including the BEAT open-science platform.

anjos.ai/projects/reproducibility

2016–2017 Remote Photoplethysmography and Vital Signs

Reproducible, camera-based estimation of vital signs, putting remote photoplethysmography on a shared, openly benchmarked footing.

anjos.ai/projects/rppg-vital-signs

2011–2019 Biometrics and Presentation-Attack Detection

Biometrics Security & Privacy Group, Idiap

Earlier work with the Idiap Biometrics group on robust recognition and presentation-attack detection that generalises to unseen attacks.

anjos.ai/projects/biometrics-pad

1998–2009 High-Energy Physics — ATLAS Trigger (CERN)

CERN, ATLAS Experiment; Federal University of Rio de Janeiro

Earlier work at CERN's ATLAS experiment on real-time trigger and data-acquisition systems, and neural-network methods for particle recognition.

anjos.ai/projects/hep-cern

GRANTS AND FUNDING

2026 – 2030 IMAGIN-AIR: From Images to Knowledge: AI to Elucidate Mechanisms and Improve Therapeutic Strategies for Retinal Inflammation

Carigest

www.idiap.ch/en/scientific-research/projects/IMAGIN-AIR

2025 – 2028 AI2Pub: AI Insights to the Public through bi-directional media channels

Swiss National Science Foundation (Instrument: Agora)

data.snf.ch/grants/grant/232373

2024 – 2028 FairMI: Machine Learning Fairness with Application to Medical Images

Swiss National Science Foundation (Instrument: Lead Agency)

- data.snf.ch/grants/grant/214653
- 2025 – 2027 **DREAM: Detection and Research for Learning Sets on Electroencephalographic Patterns**
Assistance Publique – Hôpitaux de Paris
www.idiap.ch/en/scientific-research/projects/DREAM
- 2025 – 2026 **VALIDATE-H: VALAls aDaptation of Artificial inTelligence modELs in Healthcare**
Service des Hautes Écoles
www.idiap.ch/en/scientific-research/projects/VALIDATE-H
- 2024 – 2025 **CollabCloud: A Collaborative Research Cloud Infrastructure powering Discovery and Exchange**
Swiss National Science Foundation
data.snf.ch/grants/grant/220409
- 2023 – 2024 **NewsOnAI: Scientific voice to latest news of AI technology**
Swiss National Science Foundation
data.snf.ch/grants/grant/214101
- 2023 – 2023 **Nouvelle plateforme de recherche « Cloud » pour étendre les capacités de recherche, d'échange et de formation de l'Idiap**
Loterie Romande
loro.ch/fr/beneficiaires/projets-soutenus/collabcloud
- 2022 – 2023 **CAD4IED: Computer Aided Diagnosis for Inflammatory Eye Diseases**
Idiap Research Institute
www.idiap.ch/en/scientific-research/projects/CAD4IED
- 2019 – 2023 **LEARN-REAL: LEARNING physical manipulation skills with simulators using REAListic variations**
Swiss National Science Foundation (Instrument: CHIST-ERA)
data.snf.ch/grants/grant/180417
- 2022 – 2023 **SECure: Safe and Explainable Clinical AI for Orthopaedic Surgical Assessment**
TheArk
- 2019 – 2020 **SEWS2: Smart Early Warning Score System for in- and out-hospital care via anomaly detection**
Innosuisse – Swiss Innovation Agency (Instrument: Innovation check)

EARLIER CAREER CONTRIBUTIONS

Mostly while working at the Biometrics & Security group of Sébastien Marcel.

- 2019 – 2021 **AI4EU: A European AI On Demand Platform and Ecosystem**
European Commission (Instrument: Horizon 2020)
Roles: Co-authorship of the Idiap contribution, project management
www.idiap.ch/en/scientific-research/projects/AI4EU
- 2018 – 2021 **ALLIES: Autonomous Lifelong learning intelligent Systems**
Swiss National Science Foundation (Instrument: CHIST-ERA)
Roles: Co-authorship, project management, implementation
data.snf.ch/grants/grant/174239
- 2017 – 2021 **BATL: Biometric Authentication with a Timeless Learner**
US Intelligence Advanced Research Projects Activity (IARPA) (Instrument: Odin program)
Roles: Co-authorship of the Idiap participation (face modality)
www.idiap.ch/~marcel/project/former/batl
- 2012 – 2016 **BEAT: Biometrics Evaluation and Testing**
European Commission (Instrument: Seventh Framework Programme (FP7))
Roles: Co-authorship, implementation, project management
www.idiap.ch/en/scientific-research/projects/BEAT
- 2014 – 2015 **COHFACE: Contactless Heartbeat detection for trustworthy FACE Biometrics**
Hasler Foundation
Roles: Co-authorship, dataset acquisition management, implementation
www.idiap.ch/en/scientific-research/projects/COHFACE
- 2013 – 2014 **FEDARS: Feature Extraction from Deep learning Architectures for face Recognition Systems**
Swiss Commission for Technology and Innovation (CTI)
Roles: Implementation
www.idiap.ch/en/scientific-research/projects/FEDARS
- 2010 – 2014 **TABULA RASA: Trusted Biometrics under Spoofing Attacks**
European Commission (Instrument: Seventh Framework Programme (FP7))
Roles: Dataset acquisition (WP2), implementation (WP4)
www.idiap.ch/en/scientific-research/projects/TABULARASA

TEACHING

- 2019–2023 **Master in Artificial Intelligence: Modules M05, M06 and M08**
Idiap & Unidistance
Three modules taught between 2019 and 2023 to master students of the Master in AI run by the Idiap Research Institute with Unidistance: open science and ethics (M05), and machine learning fundamentals (M06 and M08).
- 2013–2023 **Fundamentals of Statistical Pattern Recognition**
EPFL, Lausanne
EE-612 at EPFL: fundamental tools of machine learning and pattern recognition, from basics to deep learning, for Ph.D. students. Given biennially from 2013 to 2023.
- 2022 **IFI/UZH Summer School: From Scripts to Reusable Software and Reproducible Research**
University of Zurich
Summer school at the University of Zurich: actionable guidelines for turning non-reproducible scripts with hard-coded parameters into reusable software, with examples in Python.
- 2015 **Reproducible Research for Pattern Recognition**
UNESP Bauru, Brazil
A hands-on course given at UNESP Bauru, Brazil, on reproducible research for engineers working with software in pattern recognition and machine learning.

SUPERVISION

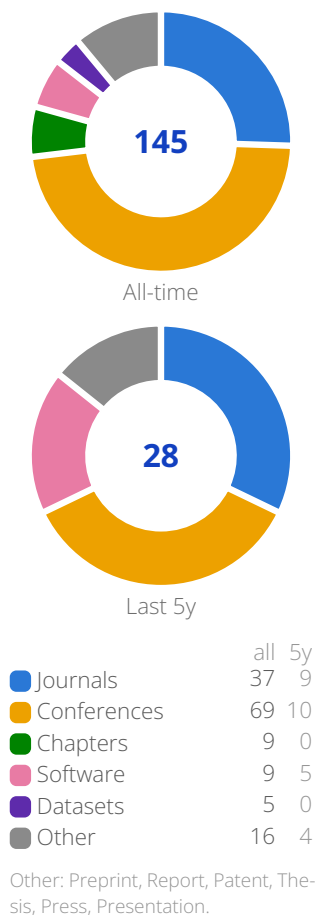
- 2026 **Tymo van Rijn**
Bachelor thesis  Hogeschool Rotterdam
Beyond the Last Frame: Temporal Deep Learning for Automated Grading of Retinal Inflammation in Fluorescein Angiography
www.idiap.ch/~aanjos/pdfs/theses/tymo-van-rijn.pdf
- 2025 **Daniel Lutziger**
Master thesis  University of Zürich, Department of Informatics
Towards Scalable Foundation Models for Sleep EEG and Polysomnography
seaf.ifi.uzh.ch/f/9951eb6cc9c64b0fa330
- 2025 **Maximilian Achakri**
Master thesis  University of Zürich, Department of Informatics
Towards Computer-Aided Diagnosis of Fibromuscular Dysplasia: Abnormality Detection in the Renal Artery Using Deep Learning
www.idiap.ch/~aanjos/pdfs/theses/maximilian-achakri.pdf
- 2025 **Reto Odoni**
Master thesis  University of Zürich, Department of Informatics
Measuring Bias in AI Anatomical Structure Segmentation Models
seaf.ifi.uzh.ch/f/da14adc2a0a14e70964d
- 2025 **Roberto Pulvirenti**
Master thesis  Politecnico di Torino, MSc in Artificial Intelligence & Data Analytics
Evaluating Cross-Domain Adaptation Strategies for Foundation Models: A Study on Fluorescein Angiography
www.idiap.ch/~aanjos/pdfs/theses/roberto-pulvirenti.pdf
- 2024 **Joël Inglin**
Master thesis  University of Zürich, Department of Informatics
Stability and Accuracy of Radiomic Features in Medical Image Classification
seaf.ifi.uzh.ch/f/4a5dbaf45523404d860d
- 2024 **Yvan Pannatier**
Master thesis  University of Geneva, Faculty of Science (MSc Computer Science)
Clinically Interpretable Computer-Aided Diagnostic Tool for Fibromuscular Dysplasia Detection
www.idiap.ch/~aanjos/pdfs/theses/yvan-pannatier.pdf
- 2023 **Gabriele Brunini**
Master thesis  University of Zürich, Department of Informatics
Automatic Sleep-Phase Analysis via Stateful Methods
- 2023 **Geoffrey Raposo**
Master thesis  UniDistance, Brig (Master in AI, jointly with Idiap)
Active Tuberculosis Detection from Frontal Chest X-ray Images
http://publications.idiap.ch/attachments/reports/2021/Raposo_Idiap-Com-01-2021.pdf
- 2023 **Özgür Güler**
Master thesis  University of Zürich
Explaining CNN-Based Active Tuberculosis Detection through Saliency Mapping
capuana.ifi.uzh.ch/publications/PDFs/24103_Master_Thesis_oezguer_acar_gueler.pdf
- 2023 **Samuel Michel**

- Master thesis  UniDistance, Brig (Master in AI, jointly with Idiap)
Generalizable Automatic Classification of Sleep Stages
publications.idiap.ch/attachments/reports/2023/Michel_Idiap-Com-02-2023.pdf
- 2023 **Victor Amiot**
Master thesis  EPFL, School of Life Sciences
Automatic Grading of Vasculitis Inflammation in Fluorescein Angiography Images
www.idiap.ch/~aanjos/pdfs/theses/victor-amiot-vasculitis.pdf
- 2023 **Xiao Tan**
Master thesis  University of Zürich, Department of Informatics
Semantic Segmentation of Weakly Labeled Retinal Images
capuana.ifi.uzh.ch/publications/PDFs/23611_Xiao%20Tan_thesis.pdf
- 2022 **Antonio Morais**
Master thesis  EPFL, Electrical Engineering
Is System A Statistically Better Than System B?
publiadiap.idiap.ch/attachments/reports/2022/Morais_Idiap-Com-01-2023.pdf
- 2022 **Driss Khalil**
Master thesis  UniDistance, Brig (Master in AI, jointly with Idiap)
Computer-Aided Screening for Eye Diseases from 2-D Retinography
www.idiap.ch/~aanjos/pdfs/theses/driss-khalil.pdf
- 2022 **Maxime Délitroz**
Master thesis  EPFL
Automated Segmentation of Developing Motor Neurons in Fluorescence Microscopy
- 2020 **Colombine Verzat**
Master thesis  UniDistance, Brig (Master in AI, jointly with Idiap)
Machine Learning for Adverse-Event Detection in Latent Tuberculosis Treatment
publications.idiap.ch/attachments/reports/2020/Verzat_Idiap-Com-02-2020.pdf
- 2015 **Ivana Chingovska**
De facto co-supervisor  EPFL, Electrical Engineering (doctoral programme, with Idiap)
Trustworthy Biometric Verification under Spoofing Attacks: Application to the Face Mode
publications.idiap.ch/attachments/papers/2016/Chingovska_THESIS_2015.pdf
- 2018 **Jaden Diefenbaugh**
Graduate student  Idiap Research Institute, Switzerland
Work associated with the BEAT platform.
- 2010 – 2013 **Tiago de Freitas Pereira and Jukka Määtä**
Visiting students, co-supervision  Idiap Research Institute, Switzerland
Presentation-attack detection in biometrics.
- 2010 – 2013 **Murali Mohan Chakka**
Trainee, co-supervision  Idiap Research Institute, Switzerland
Presentation-attack detection in biometrics.

Bibliometrics

h-index: 32
Citations: 29'000+
Google Scholar, 2026

Research Outputs



OPEN-SOURCE SOFTWARE

- 2025 **fairical**
A Python library for evaluating and tuning machine-learning systems via multi-objective optimisation, balancing fairness metrics against utility; it approximates Pareto fronts of utility-fairness trade-offs with convergence, diversity, and hypervolume scores.
gitlab.idiap.ch/medai/software/fairical
- 2024 **mednet**
A framework for developing and analysing deep neural network architectures for medical data (2D/3D images and signals), with ready access to many public medical-imaging datasets and extensible model architectures.
gitlab.idiap.ch/medai/software/mednet
- 2023 **clapper**
Configuration and command-line-interface support for Python packages: automatic TOML-based RC configuration storage/retrieval and integration with the Click framework for building CLIs.
github.com/idiap/clapper
- 2023 **credible**
scikit-learn-compatible implementations of machine-learning metrics augmented with credible/confidence intervals, so performance measurements come with quantified uncertainty bounds.
gitlab.idiap.ch/medai/software/credible
- 2023 **auto-intersphinx**
A Sphinx extension that fills intersphinx mappings from package names, removing the need to manually specify dependency documentation URLs and keeping cross-references up to date.
github.com/idiap/auto-intersphinx
- 2017 **sleepless**
A benchmarking package for evaluating machine- and deep-learning techniques on polysomnography biosignals for sleep-stage classification, covering feature extraction, training, prediction, and evaluation across datasets and algorithms.
gitlab.idiap.ch/biosignal/software/sleepless
- 2016 **BEAT**
A web-based open-science platform for running, sharing, and reusing reproducible machine-learning experiments with secure access to data and code, even when the underlying data cannot be published.
gitlab.idiap.ch/beat
- 2015 **Bob**
A free signal-processing and machine-learning toolbox developed at Idiap: a comprehensive, Python-first framework to build, experiment with, and reproduce pattern-recognition and machine-learning workflows.
gitlab.idiap.ch/bob/bob
- 2012 **gridtk**
A user-friendly command-line tool for Slurm job management — a drop-in replacement for sbatch adding local job IDs, automatic log management, monitoring, and simplified job dependencies.
github.com/idiap/gridtk

OPEN DATASETS

- 2019 **WMCA (Wide Multi-Channel presentation Attack)**
doi.org/10.34777/8zdh-v182
- 2016 **COHFACE**
doi.org/10.34777/ff3f-ba56
- 2015 **MSSpoof (Multispectral-Spoof)**
doi.org/10.34777/payf-vb10
- 2012 **Replay-Attack**
doi.org/10.34777/cwgc-7r82
- 2011 **Print-Attack**

- 2026 V. Amiot, R. Pulvirenti, O. Jimenez-del-Toro, M. Ott, T. Bogaciu, S. Banerjee, C. Amstutz, J. Odobez, C. Chiquet, Y. Guex-Crosier, C. Bergin, I. Meloni, **A. Anjos**, F. Hoogewoud, and M. Tomasoni, "UveAI: clinic-ready scoring of retinal inflammation in uveitis on widefield fluorescein angiography using AI", *Scientific reports*, 2026, doi: 10.1038/s41598-026-46069-w.
D. Q. Neto, A. Carlos, **A. Anjos**, and L. Berton, "Fair Foundation Models for Medical Image Analysis: Challenges and Perspectives", *ACM Transactions on Computing for Healthcare*, 2026, doi: 10.1145/3793542.
- 2025 G. Özbulak, O. Jimenez-del-Toro, M. Faretto, L. Berton, and **A. Anjos**, "A Multi-Objective Evaluation Framework for Analyzing Utility-Fairness Trade-Offs in Machine Learning Systems", *The Journal of Machine Learning for Biomedical Imaging* 3:938-957, 2025, doi: 10.59275/j.melba.2025-ab9a.
M. Faretto, G. Özbulak, L. Berton, and **A. Anjos**, "Optimizing Fairness and Utility in Healthcare Machine Learning Models", *Applied Soft Computing* 181(113426), 2025, doi: 10.1016/j.asoc.2025.113426.
V. Amiot, O. Jimenez-del-Toro, Y. Guex-Crosier, M. Ott, T. Bogaciu, S. Banerjee, J. Howell, C. Amstutz, C. Chiquet, C. Bergin, I. Meloni, M. Tomasoni, F. Hoogewoud, and **A. Anjos**, "Automatic transformer-based grading of multiple retinal inflammatory signs in uveitis on fluorescein angiography", *Computers in Biology and Medicine* 193:110327, 2025, doi: 10.1016/j.combiomed.2025.110327.
- 2024 T. Mautuit, P. Cunnac, F. Truffer, **A. Anjos**, R. Dufrane, G. Maître, M. Geiser, and C. Chiquet, "Absolute retinal blood flow in healthy eyes and in eyes with retinal vein occlusion", *Microvascular Research* 152:104648, 2024, doi: 10.1016/j.mvr.2023.104648.
- 2023 C. Geric, Z. Z. Qin, C. M. Denking, S. V. Kik, B. Marais, **A. Anjos**, P. David, F. A. Khan, and A. Trajman, "The rise of artificial intelligence reading of chest X-rays for enhanced TB diagnosis and elimination", *The International Journal of Tuberculosis and Lung Disease*, 2023, doi: 10.5588/ijtld.22.0687.
- 2022 M. Shamsi, A. Larcher, L. Barrault, S. Meignier, Y. Prokopalo, M. Tahon, A. Mehrish, S. Petitrenaud, O. Galibert, S. Gaist, **A. Anjos**, S. Marcel, and M. R. Costa-jussà, "Towards lifelong human assisted speaker diarization", *Computer Speech & Language*:101437, 2022, doi: 10.1016/j.csl.2022.101437.
A. Galdran, **A. Anjos**, J. Dolz, H. Chakor, H. Lombaert, and I. B. Ayed, "State-of-the-art retinal vessel segmentation with minimalistic models", *Scientific Reports* 12(1):6174, 2022, doi: 10.1038/s41598-022-09675-y.
- 2020 A. C. B. H. Ferreira, D. D. Ferreira, H. C. Oliveira, I. C. de Resende, **A. Anjos**, and M. H. B. M. Lopes, "Competitive neural layer-based method to identify people with high risk for diabetic foot", *Computers in Biology and Medicine* 120:103744, 2020, doi: 10.1016/j.combiomed.2020.103744.
- 2019 T. F. Pereira, **A. Anjos**, and S. Marcel, "Heterogeneous Face Recognition Using Domain Specific Units", *IEEE Transactions on Information Forensics and Security*, 2019, doi: 10.1109/tifs.2018.2885284.
A. George, Z. Mostaani, D. Geissenbuhler, O. Nikisins, **A. Anjos**, and S. Marcel, "Biometric Face Presentation Attack Detection with Multi-Channel Convolutional Neural Network", *IEEE Transactions on Information Forensics and Security*, 2019, doi: 10.1109/tifs.2019.2916652.
- 2016 A. Morales, J. Fierrez, R. Tolosana, J. Ortega-Garcia, J. Galbally, M. Gomez-Barrero, **A. Anjos**, and S. Marcel, "Keystroke Biometrics Ongoing Competition", *IEEE Access* 4:7736–7746, 2016, doi: 10.1109/access.2016.2626718.
- 2015 I. Chingovska, and **A. Anjos**, "On the use of client identity information for face anti-spoofing", *IEEE Transactions on Information Forensics and Security, Special Issue on Biometric Anti-spoofing* 10(4):787–796, 2015, doi: 10.1109/tifs.2015.2400392.
- 2014 T. F. Pereira, J. Komulainen, **A. Anjos**, J. M. De Martino, A. Hadid, M. Pietikainen, and S. Marcel, "Face liveness detection using dynamic texture", *EURASIP Journal on Image and Video Processing* 2014:2, 2014, doi: 10.1186/1687-5281-2014-2.
I. Chingovska, **A. Anjos**, and S. Marcel, "Biometrics Evaluation Under Spoofing Attacks", *IEEE Transactions on Information, Forensics and Security* 9(12), 2014, doi: 10.1109/tifs.2014.2349158.
- 2013 **A. Anjos**, M. M. Chakka, and S. Marcel, "Motion-Based Counter-Measures to Photo Attacks in Face Recognition", *IET Biometrics*, 2013, doi: 10.1049/iet-bmt.2012.0071.
- 2009 R. Torres, **A. Anjos**, and J. Seixas, "Automatizing the Online Filter Test Management for a General-Purpose Particle Detector", *Computer Physics Communications*, 2009, doi: 10.1016/j.cpc.2010.10.003.
- 2008 T. A. Collaboration, G. Aad, E. Abat, J. Abdallah, A. A. Abdelalim, A. Abdesselam, O. Abidinov, B. A. Abi, M. Abolins, H. Abramowicz, E. Acerbi, B. S. Acharya, R. Achenbach, M. Ackers, D. L. Adams, F. Adamyan, T. N. Addy, M. Aderholz, C. Adorisio, P. Adragna, M. Aharrouche, S. P. Ahlen, F. Ahles, A. Ahmad, H. Ahmed, G. Aielli, P. F. Åkesson, T. P. A. Åkesson, A. V. Akimov, S. M. Alam, J. Albert, S. Albrand, M. Aleksa, I. N. Aleksandrov, M. Aleppo, F. Alessandria, C. Alexa, G. Alexander, T. Alexopoulos, G. Alimonti, *et al*, "The ATLAS Experiment at the CERN Large Hadron Collider", *Journal of Instrumentation* 3(08):S08003-S08003, 2008, doi: 10.1088/1748-0221/3/08/S08003.
- 2007 T. C. Xavier, **A. Anjos**, and J. M. de Seixas, "Discriminação Neural de Partículas para um Detector Submetido a uma Alta Taxa de Eventos", *Learning and Nonlinear Models - Revista da Sociedade Brasileira de Redes Neurais (SBRN)* 4(2):79–92, 2007, doi: 10.21528/lnlm-vol4-no2-art5.
- 2006 **A. Anjos**, R. C. Torres, J. M. Seixas, B. C. Ferreira, and T. C. Xavier, "Neural triggering system operating on high resolution calorimetry information", *Nuclear Instruments and Methods in Physics Research* 559:134–138, 2006, doi: 10.1016/j.nima.2005.11.132.
A. A. A. Collaboration, "Deployment of the ATLAS High-Level Trigger", *IEEE Transactions on Nuclear Science* 53:2144–2149, 2006, <https://api.zotero.org/users/5992358/publications/items/Q66Z8T7M/file>.
A. A. A. Collaboration, "A configuration system for the ATLAS trigger", *Journal of Instrumentation, Institute of Physics Publishing and Sissa(P05004)*, 2006, <https://api.zotero.org/users/5992358/publications/items/GZP5A97A/file>.
T. A. Collaboration, "The ATLAS Data Acquisition and Trigger : concept, design and status", *Nucl. Phys. B, Proc. Suppl.* 172:178–182, 2006, <https://api.zotero.org/users/5992358/publications/items/JUS44P7I/file>.

- 2005 T. A. Collaboration, "ATLAS DataFlow: the Read-Out Subsystem, Results from Trigger and Data-Acquisition System Testbed Studies and from Modeling", *IEEE Trans. Nucl. Sciences* 53 (2006):912–917, 2005, doi: 10.1109/rtc.2005.1547446.
- T. A. Collaboration, "Overview of the High-Level Trigger Electron and Photon Selection for the ATLAS Experiment at the LHC", *IEEE Transactions Nuclear Sciences* (2005) 53:2839–2843, 2005, <https://api.zotero.org/users/5992358/publications/items/XZ42WTX3/file>.
- T. A. Collaboration, "Implementation and Performance of the Seeded Reconstruction for the ATLAS Event Filter Selection Software", *IEEE Trans. Nucl. Sciences* 53 (2007):864–869, 2005, <https://api.zotero.org/users/5992358/publications/items/6V2WJVBVP/file>.
- 2004 J. T. Baines, C. P. Bee, A. Bogaerts, M. Bosman, D. Botterill, B. Caron, **A. Anjos**, F. Etienne, S. González, K. Karr, W. Li, C. Meessen, G. Merino, A. Negri, J. L. Pinfold, P. Pinto, Z. Qian, F. Touchard, P. Werner, S. Wheeler, F. J. Wickens, W. Wiedenmann, and G. Zobernig, "An Overview of the ATLAS High-Level Trigger Dataflow and Supervision", *IEEE Transaction on Nuclear Science* 51(3):361–366, 2004, doi: 10.1109/tns.2004.828875.
- T. A. Collaboration, "ATLAS TDAQ data collection software", *IEEE Transactions on Nuclear Science* 51:585–590, 2004, <https://api.zotero.org/users/5992358/publications/items/7WQKIXU3/file>.
- T. A. Collaboration, "Studies for a common selection software environment in ATLAS : from the Level-2 Trigger to the offline reconstruction", *IEEE Transactions on Nuclear Science* 51(3):915–920, 2004, <https://api.zotero.org/users/5992358/publications/items/92TVWGMJ/file>.
- T. A. Collaboration, "Architecture of the ATLAS high level trigger event selection software", *Nucl. Instrum. Methods Phys. Res.* 518(1–2):537–541, 2004, <https://api.zotero.org/users/5992358/publications/items/UZIF4P7D/file>.
- T. A. Collaboration, "The base-line DataFlow system of the ATLAS Trigger and DAQ", *IEEE Transactions on Nuclear Science* 51(3):470–475, 2004, <https://api.zotero.org/users/5992358/publications/items/4893J5ZH/file>.
- T. A. Collaboration, "Algorithms for the ATLAS high-level trigger", *IEEE Transactions on Nuclear Science* 51(3):367–374, 2004, <https://api.zotero.org/users/5992358/publications/items/G6S27R3P/file>.
- A. A. A. Collaboration, "The Second Level Trigger of the ATLAS Experiment at CERN's LHC", *IEEE Transaction on Nuclear Science* 51(3):909–914, 2004, <https://api.zotero.org/users/5992358/publications/items/7AFJ5V4G/file>.
- 2003 **A. Anjos**, and J. Seixas, "Neural particle discrimination for triggering interesting physics channels with calorimetry data", *Nuclear Instruments And Methods In Physics Research A - Accelerators, Spectrometers, Detectors And Associated Equipment* 502:713–715, 2003, doi: 10.1016/s0168-9002(03)00553-9.
- T. A. Collaboration, "An Overview of Algorithms for the ATLAS High Level Trigger", *IEEE Transactions on Nuclear Science* 51(3 (2004)):367–374, 2003, <https://api.zotero.org/users/5992358/publications/items/746I7VIC/file>.
- 1998 J. Seixas, L. Caloba, **A. Anjos**, B. Kastrop, A. Dantas, and R. Linhares, "A neural online triggering system based on parallel processing", *IEEE Transactions on Nuclear Science* 45(4):1814–1818, 1998, doi: 10.1109/23.710942.

CONFERENCE PAPERS

- 2026 T. van Rijn, R. Pulvirenti, M. Shiri, I. Meloni, O. Jimenez-del-Toro, M. Tomasoni, F. Hoogewoud, E. Kaisari, and **A. Anjos**, "Beyond the Last Frame: Temporal Modelling of Fluorescein Angiography for Hyperfluorescence Classification", in *Ophthalmic Medical Image Analysis (OMIA), MICCAI 2026 Workshops*, 2026, <https://api.zotero.org/users/5992358/publications/items/AC37PC75/file>.
- G. Özbülak, O. Jimenez-del-Toro, L. Berton, and **A. Anjos**, "Loss-Conditioned Utility-Fairness Boundary Modeling for Medical Imaging", in *Fairness of AI in Medical Imaging (FAIMI), MICCAI 2026 Workshops*, 2026, <https://api.zotero.org/users/5992358/publications/items/25WSH8DP/file>.
- R. Pulvirenti, O. Jimenez-del-Toro, M. Tomasoni, F. Hoogewoud, and **A. Anjos**, "When Specialization Helps (and Hurts): Cross-Modality Transfer in Ophthalmic Imaging with Foundation Models", in *2026 IEEE 23rd International Symposium on Biomedical Imaging*, 2026, doi: 10.1109/isbi61048.2026.11515636.
- 2025 D. Geissbuhler, A. Bornet, C. Marques, **A. Anjos**, S. Pereira, and D. Teodoro, "STM-GNN: Space-Time-and-Memory Graph Neural Networks for Predicting Multi-Drug Resistance Risks in Dynamic Patient Networks", in *International Conference on Artificial Intelligence in Medicine*, 2025, doi: 10.1007/978-3-031-95838-0_16.
- M. Vitek, D. Tomašević, A. Das, S. Nathan, G. Özbülak, G. A. T. Özbülak, J. Calbimonte, **A. Anjos**, H. H. Bhatt, D. D. Premani, J. Chaudhari, C. Wang, J. Jiang, C. Zhang, Q. Zhang, I. I. Ganapathi, S. S. Ali, D. Velayudan, M. Assefa, N. Werghi, Z. A. Daniels, L. John, R. Vyas, J. N. Khirak, T. A. Saeed, M. Nasehi, A. Kianfar, M. P. Panahi, G. Sharma, P. R. Panth, R. Ramachandra, A. Nigam, U. Pal, H. Pedrini, and V. Struc, "Privacy-enhancing Sclera Segmentation Benchmarking Competition: SSBC 2025", in *2025 IEEE International Joint Conference on Biometrics (IJCB):1-13*, 2025, doi: 10.1109/ijcb65343.2025.11411539.
- 2024 Ö. A. Güler, M. Günther, and **A. Anjos**, "Refining Tuberculosis Detection in CXR Imaging: Addressing Bias in Deep Neural Networks via Interpretability", in *2024 12th European Workshop on Visual Information Processing (EUVIP):1-6*, 2024, doi: 10.1109/euvip61797.2024.10772829.
- D. Queiroz, **A. Anjos**, and L. Berton, "Using Backbone Foundation Model for Evaluating Fairness in Chest Radiography Without Demographic Data", in *Ethics and Fairness in Medical Imaging:109-118*, 2024, doi: 10.1007/978-3-031-72787-0_11.
- D. Queiroz, A. Carlos, M. Faretto, L. F. Nakayama, **A. Anjos**, and L. Berton, "Does Data-Efficient Generalization Exacerbate Bias in Foundation Models?", in *Fairness and Ethics Towards Transparent AI: Facing the Challenge through Model Debiasing (FAILED), ECCV 2024 Workshops*, 2024, doi: 10.48550/arxiv.2408.16154.
- O. Jimenez-del-Toro, C. Aberle, R. Schaer, M. Bach, K. Flouris, E. Konukoglu, B. Stieltjes, M. M. Obmann, **A. Anjos**, H. Müller, and A. Depeursinge, "Comparing Stability and Discriminatory Power of Hand-crafted Versus Deep Radiomics: A 3D-Printed Anthropomorphic Phantom Study", in *2024 12th European Workshop on Visual Information Processing (EUVIP)*, 2024, doi: 10.1109/euvip61797.2024.10772813.

- 2023 V. Amiot, O. Jimenez-Del-Toro, P. Eyraud, Y. Guex-Crosier, C. Bergin, **A. Anjos**, F. Hoogewoud, and M. Tomasoni, "Fully Automatic Grading of Retinal Vasculitis on Fluorescein Angiography Time-lapse from Rheumatologic Uveitis", in *2023 IEEE 36th International Symposium on Computer-Based Medical Systems (CBMS)*, 2023, doi: 10.1109/cbms58004.2023.00301.
- 2021 M. A. Renzo, N. Fernandez, A. A. Baceti, N. N. M. Junior, and **A. Anjos**, "Development of a lung segmentation algorithm for analog imaged chest X-Ray: preliminary results", in *Anais do 15. Congresso Brasileiro de Inteligência Computacional:1-8*, 2021, doi: 10.21528/cbic2021-123.
- 2018 O. Nikisins, T. Eglitis, **A. Anjos**, and S. Marcel, "Fast cross-correlation based wrist vein recognition algorithm with rotation and translation compensation", in *Sixth International Workshop on Biometrics and Forensics*, 2018, doi: 10.1109/ibwf.2018.8401550.
O. Nikisins, A. Mohammadi, **A. Anjos**, and S. Marcel, "On Effectiveness of Anomaly Detection Approaches against Unseen Presentation Attacks in Face Anti-Spoofing", in *The 11th IAPR International Conference on Biometrics (ICB 2018)*, 2018, doi: 10.1109/icb2018.2018.00022.
- 2017 M. Cernak, A. Komaty, A. Mohammadi, **A. Anjos**, and S. Marcel, "Bob Speaks Kaldi", in *Proceedings of Interspeech*, 2017, <https://api.zotero.org/users/5992358/publications/items/5RRHG8TA/file>.
A. Anjos, M. Günther, T. F. Pereira, P. Korshunov, A. Mohammadi, and S. Marcel, "Continuously Reproducing Toolchains in Pattern Recognition and Machine Learning Experiments", in *Thirty-fourth International Conference on Machine Learning*, 2017, <https://api.zotero.org/users/5992358/publications/items/R55HB7JQ/file>.
A. Anjos, L. E. Shafey, and S. Marcel, "BEAT: An Open-Science Web Platform", in *Thirty-fourth International Conference on Machine Learning*, 2017, <https://api.zotero.org/users/5992358/publications/items/ZQ5WW5UB/file>.
- 2013 I. Chingovska, **A. Anjos**, and S. Marcel, "Anti-spoofing in action: joint operation with a verification system", in *Computer Vision and Pattern Recognition Conference - Biometrics Workshop*, 2013, doi: 10.1109/cvprw.2013.22.
J. Komulainen, A. Hadid, M. Pietikäinen, **A. Anjos**, and S. Marcel, "Complementary Countermeasures for Detecting Scenic Face Spoofing Attacks", in *International Conference on Biometrics 2013*, 2013, doi: 10.1109/icb.2013.6612968.
T. F. Pereira, **A. Anjos**, J. M. De Martino, and S. Marcel, "Can face anti-spoofing countermeasures work in a real world scenario?", in *International Conference on Biometrics 2013*, 2013, doi: 10.1109/icb.2013.6612981.
I. Chingovska, J. Yang, Z. Lei, D. Yi, S. Z. Li, O. Kähm, C. Glaser, N. Damer, A. Kuijper, A. Nouak, J. Komulainen, T. Pereira, S. Gupta, S. Khandelwal, S. Bansal, A. Rai, T. Krishna, D. Goyal, M. Waris, H. Zhang, I. Ahmad, S. Kiranyaz, M. Gabbouj, R. Tronci, M. Pili, N. Sirena, F. Roli, J. Galbally, J. Fierrez, A. Pinto, H. Pedrini, W. S. Schwartz, A. Rocha, **A. Anjos**, and S. Marcel, "The 2nd Competition on Counter Measures to 2D Face Spoofing Attacks", in *International Conference on Biometrics 2013*, 2013, doi: 10.1109/icb.2013.6613026.
- 2012 I. Chingovska, **A. Anjos**, and S. Marcel, "On the Effectiveness of Local Binary Patterns in Face Anti-spoofing", in *IEEE International Conference of the Biometrics Special Interest Group*, 2012, <https://api.zotero.org/users/5992358/publications/items/HCXQE52H/file>.
A. Anjos, L. E. Shafey, R. Wallace, M. Günther, C. McCool, and S. Marcel, "Bob: a free signal processing and machine learning toolbox for researchers", in *ACM Multimedia 2012:1449-1452*, 2012, doi: 10.1145/2393347.2396517.
T. F. Pereira, **A. Anjos**, J. M. De Martino, and S. Marcel, "LBP-TOP based countermeasure against facial spoofing attacks", in *International Workshop on Computer Vision With Local Binary Pattern Variants*, 2012, doi: 10.1007/978-3-642-37410-4_11.
- 2011 **A. Anjos**, and S. Marcel, "Counter-Measures to Photo Attacks in Face Recognition: a public database and a baseline", in *International Joint Conference on Biometrics 2011*, 2011, doi: 10.1109/ijcb.2011.6117503.
M. M. Chakka, **A. Anjos**, S. Marcel, and others, "Competition on Counter Measures to 2-D Facial Spoofing Attacks", in *International Joint Conference on Biometrics 2011*, 2011, doi: 10.1109/ijcb.2011.6117509.
- 2009 T. A. Collaboration, "ATLAS Trigger and Data Acquisition: capabilities and commissioning", in *11th Pisa Meeting on Advanced Detectors 617:306-309*, 2009, doi: 10.1016/j.nima.2009.06.114.
T. A. Collaboration, "Configuration and Control of the ATLAS Trigger and Data Acquisition", in *The 1st international conference on Technology and Instrumentation in Particle Physics*, 2009, <https://api.zotero.org/users/5992358/publications/items/35UTZG93/file>.
T. A. Collaboration, "Atlas trigger for first physics and beyond", in *Physics at LHC 2008 29 September - October 4, 2008 Split, Croatia*, 2009, <https://api.zotero.org/users/5992358/publications/items/2PZ994WK/file>.
T. A. Collaboration, "ATLAS Trigger Status and Results From Commissioning Operations", in *Advanced Computing on High-Energy Physics 2008, Erice, Sicily, Italy*, 2009, <https://api.zotero.org/users/5992358/publications/items/DFDAJJ4/file>.
T. A. Collaboration, "Commissioning of the ATLAS High Level Trigger with Single Beam and Cosmic Rays", in *Computing in High Energy and Nuclear Physics, Prague, Czech Republic, 21 - 27 Mar 2009*, 2009, <https://api.zotero.org/users/5992358/publications/items/5FI6WSB5/file>.
T. A. Collaboration, "The ATLAS online High Level Trigger framework: experience reusing offline software components in the ATLAS trigger", in *Computing in High Energy and Nuclear Physics, Prague, Czech Republic, 21 - 27 Mar 2009*, 2009, <https://api.zotero.org/users/5992358/publications/items/HBJRN7GR/file>.
- 2008 **A. Anjos**, "Trigger Systems", in *Experimental High-Energy Physics and Associated Technologies Workshop*, 2008.
A. A. A. Collaboration, "The DAQ/HLT system of the ATLAS experiment", in *International Workshop on Advanced Computing and Analysis Techniques in Physics Research*, 2008, <https://api.zotero.org/users/5992358/publications/items/SQENSHCA/file>.
T. A. Collaboration, "Readiness of the ATLAS Trigger and Data Acquisition system for the first LHC beams", in *11th Topical Seminar On Innovative Particle And Radiation Detectors, Siena, Italy*, 2008, <https://api.zotero.org/users/5992358/publications/items/DBSTJGDP/file>.

- 2007 T. A. Collaboration, "Alignment data streams for the ATLAS Inner Detector", in *Computing for High-Energy Physics*, 2007, doi: 10.1088/1742-6596/119/2/022030.
- A. A. Collaboration, "The Configuration System of the ATLAS Trigger", in *IEEE Real-time conference*, 2007, <https://api.zotero.org/users/5992358/publications/items/5X23ZBK7/file>.
- T. A. Collaboration, "Performance of the final Event Builder for the ATLAS Experiment", in *15th IEEE Real Time Conference 2007*, 2007, <https://api.zotero.org/users/5992358/publications/items/K3M5MUZG/file>.
- T. A. Collaboration, "Integration of the Trigger and Data Acquisition systems in ATLAS", in *IEEE Real-time conference*, 2007, <https://api.zotero.org/users/5992358/publications/items/8E888TIM/file>.
- T. A. Collaboration, "The ATLAS trigger - high-level trigger commissioning and operation during early data taking", in *International Europhysics Conference on High Energy Physics*, 2007, <https://api.zotero.org/users/5992358/publications/items/KRZ7KPS2/file>.
- T. A. Collaboration, "The ATLAS Trigger - Commissioning with cosmic rays", in *International Conference on Computing in High Energy and Nuclear Physics*, 2007, <https://api.zotero.org/users/5992358/publications/items/UC7BVECC/file>.
- R. C. Torres, J. M. Seixas, **A. Anjos**, and D. V. Cunha, "Online Electron/Jet Neural High-Level Trigger over Independent Calorimetry Information", in *XI International Workshop on Advanced Computing and Analysis Techniques in Physics Research*, 2007, <https://api.zotero.org/users/5992358/publications/items/2VHMDEJM/file>.
- T. A. Collaboration, "The ATLAS Event Builder", in *IEEE Nuclear Science Symposium and Medical Imaging Conference*, 2007, <https://api.zotero.org/users/5992358/publications/items/PWF3I3E7/file>.
- 2006 T. A. Collaboration, "Testing on a Large Scale: running the ATLAS Data Acquisition and High Level Trigger Software on 700 PC Nodes", in *Computing In High Energy and Nuclear Physics*, 2006, <https://api.zotero.org/users/5992358/publications/items/753XBFJ3/file>.
- T. A. Collaboration, "Studies with the ATLAS Trigger and Data Acquisition Pre-Series Setup", in *15th International Conference on Computing In High Energy and Nuclear Physics*, 2006, <https://api.zotero.org/users/5992358/publications/items/SWU3XHMJQ/file>.
- T. A. Collaboration, "ATLAS High Level Trigger Infrastructure, ROI Collection and Event Building", in *15th International Conference on Computing In High Energy and Nuclear Physics*, 2006, <https://api.zotero.org/users/5992358/publications/items/Q5FXBHXA/file>.
- 2005 A. A. Collaboration, "Configuration of the ATLAS trigger", in *14th IEEE NPSS Real Time Conference:990-994*, 2005, <https://api.zotero.org/users/5992358/publications/items/4AIDEPTG/file>.
- A. Anjos**, R. C. Torres, B. C. Ferreira, T. C. Xavier, J. M. Seixas, and D. O. Damazio, "Otimização do Sistema de Trigger do Segundo Nível do ATLAS Baseado em Calorimetria", in *XXVI Encontro Nacional de Física de Partículas e Campos*, 2005, <https://api.zotero.org/users/5992358/publications/items/FIZVWDGQ/file>.
- T. A. Collaboration, "Muon Reconstruction and Identification for the Event Filter of the ATLAS experiment", in *9th ICATAPP Conference on High Energy Physics*, 2005, <https://api.zotero.org/users/5992358/publications/items/7W4K2GCW/file>.
- T. A. Collaboration, "Implementation and performance of a tau lepton selection within the ATLAS trigger system at the LHC", in *9th ICATPP Conference on Astroparticle, Particle, Space Physics, Detectors and Medical Physics Applications*, 2005, <https://api.zotero.org/users/5992358/publications/items/BVRS98CP/file>.
- A. Anjos**, R. C. Torres, B. C. Ferreira, T. C. Xavier, and J. M. de Seixas, "Discriminação Neural de Elétrons no Segundo Nível de Trigger do ATLAS", in *XXVI Encontro Nacional de Física de Partículas e Campos*, 2005, <https://api.zotero.org/users/5992358/publications/items/KQW3EJ5W/file>.
- 2004 **A. Anjos**, and J. M. Seixas, "Os Filtros de Alto Nível do Experimento ATLAS", in *XXVI Encontro Nacional de Física de Partículas e Campos*, 2004, <https://api.zotero.org/users/5992358/publications/items/EZT4G27A/file>.
- T. A. Collaboration, "Performance of the ATLAS DAQ DataFlow system", in *Computing in High Energy Physics and Nuclear Physics*, 2004, doi: 10.5170/cern-2005-002.91.
- T. A. Collaboration, "Online Muon Reconstruction in the ATLAS Level-2 trigger system", in *Nuclear Science Symposium and Medical Imaging Conference*, 2004, <https://api.zotero.org/users/5992358/publications/items/VBVZT92N/file>.
- T. A. Collaboration, "Portable Gathering System for Monitoring and Online Calibration at ATLAS", in *Computing in High Energy Physics and Nuclear Physics 2004*, 2004, <https://api.zotero.org/users/5992358/publications/items/76ICCDF8/file>.
- T. A. Collaboration, "Implementation and Performance of the High Level Trigger Electron and Photon Selection for the ATLAS Experiment at the LHC", in *IEEE Nuclear Science Symposium and Medical Imaging Conference*, 2004, <https://api.zotero.org/users/5992358/publications/items/V74FGXF7/file>.
- T. A. Collaboration, "Design, deployment and functional tests of the on-line Event Filter for the ATLAS experiment at LHC", in *Nuclear Science Symposium and Medical Imaging Conference*, 2004, <https://api.zotero.org/users/5992358/publications/items/FZHVD6CE/file>.
- 2003 T. A. Collaboration, "Experience with multi-threaded C++ applications in the ATLAS dataflow software", in *Conference for Computing in High-Energy and Nuclear Physics*, 2003, <https://api.zotero.org/users/5992358/publications/items/C5H5GAMD/file>.
- T. A. Collaboration, "A New Implementation of the Region-of-Interest Strategy for the ATLAS Second Level Trigger", in *Conference for Computing in High-Energy and Nuclear Physics*, 2003, <https://api.zotero.org/users/5992358/publications/items/2EFDCRZJ/file>.
- T. A. Collaboration, "The Algorithm Steering and Trigger Decision mechanism of the ATLAS High Level Trigger", in *Conference for Computing in High-Energy and Nuclear Physics*, 2003, <https://api.zotero.org/users/5992358/publications/items/GAGP9ESQ/file>.

- T. A. Collaboration, "Architecture of the ATLAS online physics-selection software at LHC", in *Conference on Astroparticle, Particle, Space Physics, Detectors and Medical Physics Applications*, 2003, <https://api.zotero.org/users/5992358/publications/items/QW5QZ3G5/file>.
- T. A. Collaboration, "The baseline dataflow system of the ATLAS trigger and DAQ", in *9th Workshop on Electronics for LHC Experiments*, 2003, <https://api.zotero.org/users/5992358/publications/items/J3C6B9JF/file>.
- T. A. Collaboration, "The DataFlow System of the ATLAS Trigger and DAQ", in *Conference for Computing in High-Energy and Nuclear Physics*, 2003, <https://api.zotero.org/users/5992358/publications/items/ETM97WMJ/file>.
- 2001 **A. Anjos**, and J. Seixas, "Redes Neurais especialistas para a separação Elétron-Jato usando Calorímetros multi-camadas e multi-segmentados", in *XXII Encontro Nacional de Física de Partículas e Campos*, 2001, <https://api.zotero.org/users/5992358/publications/items/47S2VMRK/file>.
- 2000 **A. Anjos**, and J. M. de Seixas, "Mapeamento em anéis para uma separação neuronal elétron-jato usando calorímetros multi-camadas e multi-segmentados", in *XIX Encontro Nacional de Física de Partículas e Campos*, 2000, <https://api.zotero.org/users/5992358/publications/items/PIQHGV99/file>.
- 1999 **A. Anjos**, and J. M. de Seixas, "Integrando Plataformas e Algoritmos para o Segundo Nível de Trigger do Experimento ATLAS", in *Encontro Nacional de Física de Partículas e Campos*, 1999, <https://api.zotero.org/users/5992358/publications/items/ADE5P7D9/file>.
- 1998 **A. Anjos**, A. Dantas, and J. M. de Seixas, "Um Protótipo do Sistema de Validação do Nível 2 para as Condições do LHC", in *Encontro Nacional de Física de Partículas e Campos*:32–33, 1998, <https://api.zotero.org/users/5992358/publications/items/AE3ZBAWP/file>.
- J. M. Seixas, **A. Anjos**, C. B. Prado, L. P. Calôba, A. C. H. Dantas, and J. C. R. Aguiar, "Neural classifiers implemented in a transputer based parallel machine", in *International Meeting on Vector and Parallel Processing (VECPAR)*, 1998, <https://api.zotero.org/users/5992358/publications/items/9VMFH959/file>.
- 1997 J. Seixas, L. Calôba, **A. Anjos**, A. Dantas, and R. Linhares, "Fast Neural Decision System Based On DSPs And Parallel Processing", in *International Conference on Signal Processing Applications and Technologies, San Diego, USA*:1629–1633, 1997, <https://api.zotero.org/users/5992358/publications/items/XMCRWBKF/file>.
- 1996 J. Seixas, L. Calôba, and **A. Anjos**, "Particle discrimination using sub-optimal filtering techniques", in *Congresso Brasileiro de Automatica (CBA), São Paulo, Brasil*:635–640, 1996, <https://api.zotero.org/users/5992358/publications/items/KMMH9868/file>.

BOOK CHAPTERS

- 2019 S. Bhattacharjee, A. Mohammadi, **A. Anjos**, and S. Marcel, "Recent Advances in Face Presentation Attack Detection", *Handbook of Biometric Anti-Spoofing*:207–228, 2019, doi: 10.1007/978-3-319-92627-8_10.
- A. Anjos**, P. Tome, and S. Marcel, "An Introduction to Vein Presentation Attacks and Detection", *Handbook of Biometric Anti-Spoofing*:419–438, 2019, doi: 10.1007/978-3-319-92627-8_18.
- I. Chingovska, A. Mohammadi, **A. Anjos**, and S. Marcel, "Evaluation Methodologies for Biometric Presentation Attack Detection", *Handbook of Biometric Anti-Spoofing*:457–480, 2019, doi: 10.1007/978-3-319-92627-8_20.
- 2016 I. Chingovska, N. Erdogmus, **A. Anjos**, and S. Marcel, "Face Recognition Systems Under Spoofing Attacks", *Face Recognition Systems Under Spoofing Attacks*:165–194, 2016, doi: 10.1007/978-3-319-28501-6_8.
- 2014 S. Z. Li, J. Galbally, **A. Anjos**, and S. Marcel, "Evaluation Databases", *Handbook of Biometric Anti-Spoofing*:247–278, 2014, doi: 10.1007/978-1-4471-6524-8.
- A. Anjos**, J. Komulainen, S. Marcel, A. Hadid, and M. Pietikainen, "Face Anti-Spoofing: Visual Approach", *Handbook of Biometric Anti-Spoofing*:65–82, 2014, doi: 10.1007/978-1-4471-6524-8_4.
- I. Chingovska, **A. Anjos**, and S. Marcel, "Evaluation Methodologies", *Handbook of Biometric Anti-Spoofing*:185–204, 2014, doi: 10.1007/978-1-4471-6524-8_10.
- I. Chingovska, **A. Anjos**, and S. Marcel, "Anti-spoofing: Evaluation Methodologies", *Encyclopedia of Biometrics*, 2014, doi: 10.1007/978-3-642-27733-7.
- A. Anjos**, I. Chingovska, and S. Marcel, "Anti-Spoofing: Face Databases", *Encyclopedia of Biometrics*, 2014, doi: 10.1007/978-3-642-27733-7_9212-2.

PREPRINTS

- 2025 D. Geissbuhler, A. Bornet, C. Marques, **A. Anjos**, S. Pereira, and D. Teodoro, "STM-GNN: Space-Time-and-Memory Graph Neural Networks for Predicting Multi-Drug Resistance Risks in Dynamic Patient Networks", *Health Informatics*, 2025, doi: 10.1101/2025.05.27.25327491.
- 2024 V. Amiot, O. Jimenez-del-Toro, Y. Guex-Croisier, M. Ott, T. Bogaciu, S. Banerjee, J. Howell, C. Amstutz, C. Chiquet, C. Bergin, I. Meloni, M. Tomasoni, F. Hoogewoud, and **A. Anjos**, "Automatic Transformer-Based Grading of Multiple Retinal Inflammatory Signs on Fluorescein Angiography", *SSRN*, 2024, doi: 10.2139/ssrn.4960069.
- 2020 A. Galdran, **A. Anjos**, J. Dolz, H. Chakor, H. Lombaert, and I. B. Ayed, "The Little W-Net That Could: State-of-the-Art Retinal Vessel Segmentation with Minimalistic Models", *arXiv*, 2020, doi: 10.48550/arxiv.2009.01907.
- T. Laibacher, and **A. Anjos**, "On the Evaluation and Real-World Usage Scenarios of Deep Vessel Segmentation for Retinography", *arXiv*, 2020, doi: 10.48550/arxiv.1909.03856.
- 2017 **A. Anjos**, L. El-Shafey, and S. Marcel, "BEAT: An Open-Source Web-Based Open-Science Platform", *arXiv*, 2017, doi: 10.48550/arxiv.1704.02319.
- G. Heusch, **A. Anjos**, and S. Marcel, "A reproducible study on remote heart rate measurement", *arXiv*, 2017, doi: 10.48550/arxiv.1709.00962.

TECHNICAL REPORTS

- 2008 T. A. Collaboration, "Expected Performance of the ATLAS Experiment Detector, Trigger, Physics", *CERN Open Documentation:2008-020*, 2008, doi: 10.48550/arxiv.0901.0512.
- 2003 T. A. Collaboration, "The ATLAS HLT, DAQ and DCS Technical Design Report", *CERN Publication*, 2003, <https://api.zotero.org/users/5992358/publications/items/53DHC686/file>.

PATENTS

- 2019 L. S. Momo, L. C. Torres, S. Marcel, **A. Anjos**, M. Liebling, A. Shajkofci, S. Amoos, A. Woeffray, A. Sierro, P. Roduit, P. Ferrez, and L. Bonvin, "Method and Device for Biometric Vascular Recognition and/or Identification", *WO/2019/150254*, 2019, <https://api.zotero.org/users/5992358/publications/items/QG2BP7FM/file>.
- 2018 M. Sébastien, **A. Anjos**, and P. Abbet, "Method and internet-connected server for reviewing a computer-executable experiment", *US9973503B2*, 2018, <https://api.zotero.org/users/5992358/publications/items/A2JXK46I/file>.
- 2017 **A. Anjos**, and S. Marcel, "A data-network connected server, a device, a platform and a method for conducting computer-executable experiments", *WO/2017/221049*, 2017, <https://api.zotero.org/users/5992358/publications/items/93HG875F/file>.

THESES

- 2006 **A. Anjos**, "Sistema Online de Filtragem em um Ambiente com Alta Taxa de Eventos", 2006, <https://api.zotero.org/users/5992358/publications/items/6SWJXRDZ/file>.
- 2001 **A. Anjos**, "Sistema neuronal rápido de decisão baseado em calorimetria de altas energias", 2001, <https://api.zotero.org/users/5992358/publications/items/C8FM5BTA/file>.
- 1997 **A. Anjos**, "Sistema de classificação baseado em uma máquina com sistema distribuído", 1997, <https://api.zotero.org/users/5992358/publications/items/3MENIXQM/file>.